

**AN OLD-NEW CONCEPT OF CONVEX RISK MEASURES:
THE OPTIMIZED CERTAINTY EQUIVALENT**

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Abstract

The optimized certainty equivalent (OCE) is a decision theoretic criterion based on a utility function, that was first introduced by the authors in 1986. This paper re-examines this fundamental concept, studies and extends its main properties, and put it in perspective to recent concepts of risk measures. We show that the negative of the OCE naturally provides a wide family of risk measures that fits the axiomatic formalism of convex risk measures. Duality theory is used to reveal the link between the OCE and the φ -divergence functional (a generalization of relative entropy), and allows for deriving various variational formulas for risk measures. Within this interpretation of the OCE, we prove that several risk measures recently analyzed and proposed in the literature (e.g., conditional value of risk, bounded shortfall risk) can be derived as special cases of the OCE by using particular utility functions. We further study the relations between the OCE and other certainty equivalents, providing general conditions under which these can be viewed as coherent/convex risk measures. Throughout the paper several examples illustrate the flexibility and adequacy of the OCE for building risk measures.

Key Words : Expected Utility, Certainty Equivalents, Coherent Risk Measures, Convex Risk Measures, Risk Aversion, Convex Duality, Information Theory, φ -divergences, Decision Making under Uncertainty, Conditional Value at Risk, Penalty Functions, Shortfall Risk.

1 Introduction

A major concern in finance is how to quantify the risk associated with future random outcomes, such as interest rates, stock prices, exchange rates etc. In the pioneering work of Markovitz [21], the risk measure of a random monetary outcome was its variance. In recent years however, in particular after the publication of the seminal paper by Artzner, Delbaen, Eber and Heath [2] the focus is on risk measures satisfying certain acceptable properties ("axioms"), some of which are not shared by the variance. In the work of Artzner et al. [2], the properties imposed on a risk measure $\rho(X)$ of a random variable X , is that ρ is a real-valued function which is positively homogeneous, subadditive, translation invariant and monotone. Such risk measures are called *coherent*. After the introduction of coherent risk measures, many variations and extensions of them have been proposed and studied in the literature. To cite just a few, see for example the works of [1], [15], [23], [24], [27], [28], and references therein. An important extension of the concept of coherent risk measure is the notion of *convex risk measure*, developed in the recent works of Föllmer and Shied [15], and Frittelli and Gianin [14], who proposed to extend the notion of coherent risk measures by relaxing the subadditivity and positive homogeneity conditions on ρ with the weaker property of convexity:

$$\rho(\lambda X + (1 - \lambda)Y) \leq \lambda \rho(X) + (1 - \lambda)\rho(Y), \quad \forall \lambda \in [0, 1].$$

Convexity implies that diversification does not increase the risk, and thus appears as a viable property for a risk measure.

A risk measure $\rho(\cdot)$ imposes a *preference order* $\succeq$ on random variables, namely

$$X \succeq Y \text{ if and only if } \rho(X) \leq \rho(Y).$$

Years before the emergence of risk measures, preference orders on random variables were introduced as a result of the celebrated Expected Utility (EU) theory of von Neumann and Morgenstern, [22]. For an individual with utility function $u(\cdot)$, the preference order under the EU theory is

$$X \succeq Y \text{ if and only if } \mathbb{E}u(X) \geq \mathbb{E}u(Y),$$

or equivalently, when u is strictly increasing, (which amounts to ask the evident requirement that *more is better*),

$$X \succeq Y \text{ if and only if } C_u(X) \geq C_u(Y),$$

where, for any random variable Z , the quantity

$$C_u(Z) := u^{-1}\mathbb{E}u(Z),$$

is its *certainty equivalent*, i.e., the sure amount for which a decision maker remains indifferent to a lottery Z . Another certainty equivalent based on utility functions is the so-called *u-mean*, $M_u(\cdot)$, proposed by Bühlmann in [12], and which is defined for any random variable X by $M_u(\cdot)$ satisfying

$$\mathbb{E}u(X - M_u(X)) = 0,$$

the last equation being also known as the *principle of zero utility*.

In 1986, the authors introduced in [6] another certainty equivalent for the random outcome X based on utility functions, called here the *Optimized Certainty Equivalent* (OCE), and defined by

$$S_u(X) = \sup_{\eta \in \mathbb{R}} \{\eta + \mathbb{E}u(X - \eta)\},$$

where u is a normalized concave utility function, in which case $\mathbb{E}u(X)$ is interpreted as the sure present value of a future uncertain income X . The rational behind the definition of the optimized certainty equivalent is as follows: suppose a decision maker expects a future uncertain income of X dollars, and can consume part of X at present. If he chooses to consume η dollars, the resulting present value of X is then $\eta + \mathbb{E}u(X - \eta)$. Thus, the sure (present) value of X , (i.e., its certainty equivalent $S_u(X)$) is the result of an *optimal* allocation of X between present and future consumption. The latter also motivates the name OCE.

It happens that any certainty equivalent $CE(\cdot)$ inducing a preference order on random variables generates a corresponding risk measure $\rho(\cdot)$, inducing the same preference order by the simple relation,

$$\rho(X) = -CE(X),$$

and vice versa. As an example, the u -mean $M_u(\cdot)$ is closely related to the modern risk measure called *shortfall risk* introduced by Föllmer and Schied [15], [16], and defined by,

$$\rho^{FS}(X) = \inf\{\eta \mid \mathbb{E}u(X - \eta) \geq 0\}.$$

In fact, for a strictly increasing utility, $\rho^{FS}(X) = -M_u(X)$.

In this paper we re-examine the OCE, study and extend its main properties and put it in perspective of recent risk measure theory. We show that

$$\rho_u(X) := -S_u(X),$$

is under reasonable conditions on the utility function u , a convex risk measure. This is developed in Section 2 where we give the main properties of the OCE, and some examples of risk measures that can be derived with special choices of the utility function. In particular, we show that the minimization formula for conditional value at risk, CVAR, recently proposed by Rockafellar and Uryasev in [27] is just a special case of the OCE that corresponds to choosing the utility function as a simple piecewise linear one. In Section 3, we further fully characterize the class of utility functions for which $\rho_u(X)$ is a coherent risk measure. The latter class consists of the piecewise linear utilities parameterized by $0 \leq \gamma_1 < 1 < \gamma_2$; specifically,

$$u(t) = \gamma_1[t]_+ - \gamma_2[-t]_+, \quad ([t]_+ := \max\{0, t\}).$$

The corresponding risk measure is directly related to a risk measure recently considered by Pflug and Ruszczyński [24], and reduces to CVAR when $\gamma_1 = 0$.

Outside of Economics and Finance, *Information Theory*, is a major field dealing with models involving uncertainty. The main mathematical entities in information theory are *entropy*, *relative (or cross) entropy*, [19], and the more general concept of φ -divergence, introduced by Csiszár, [13], and defined by

$$I_\varphi(Q, P) := \int_{\Omega} \varphi\left(\frac{dQ}{dP}\right) dP,$$

where Q is a probability measure which is absolutely continuous with respect to P on some probability space $(\Omega, \mathcal{F}, P)$, and φ is a given convex function. These “entropy measures” quantify “distance” between random variables.

It turns out that the OCE is directly related to the concept of φ -divergence; in fact, the OCE originally emerged from the study of penalty functions for stochastic constraints which are constructed via φ -divergence distances, see Ben-Tal and Teboulle [7]. As shown in [7], and its sequel [9], convex duality theory is the key to establish a one-to-one correspondence between the risk measure $-S_u(X)$ and the information measure defined in terms of a φ -divergence. In Section 4 we recapture these results in a more general setting. Our results provides concrete instances of the representation theorems developed by Föllmer and Schied [15], [16] for general convex risk measures. As an application, we also show that the formula for CVAR for discrete and continuous distributions can be computed as a special case of our dual representation for OCE.

In the last section, we further study the risk measures associated with the other certainty equivalents mentioned above, namely the classical certainty equivalent $C_u(\cdot)$ and the u-mean $M_u(\cdot)$, characterizing the utility functions for which these are coherent or convex risk measures. We reveal the following simple connection between the OCE and the shortfall risk measure $\rho^{FS}(X)$ of Föllmer and Schied [15], via:

$$\rho^{FS}(X) = - \inf_{\lambda > 0} S_{\lambda u}(X).$$

This enables one to deduce properties of shortfall risk from those of OCE, such as convexity, coherence and more. In particular, the dual representation formula for OCE generates straightforwardly a corresponding formula of ρ^{FS} , which coincides with a result of Föllmer and Schied [15, Theorem 10].

The OCE, which is the central concept in this paper, was applied to analyze several economic decision problems under uncertainty. It was proved to be a powerful analytical tool for this purpose, recapturing many results obtained previously for the classical certainty equivalent $C_u(X)$, and in many cases producing results with significantly less stringent assumptions on the utility functions. We refer the interested reader to the work of Ben-Tal and Teboulle [8], where the OCE was applied to portfolio theory, and Ben-Tal and Ben-Israel [5], [10], where the OCE was applied to production, investment, inventory and insurance problems.

2 The Optimized Certainty Equivalent

In this section we give the formal definition of the Optimized Certainty Equivalent (OCE) which was first introduced by the authors in [6]. We establish its main properties, and show how it can be used to generate convex risk measures.

2.1 Main Properties of the OCE

Let Ω be a set of possible scenarios, and $(\Omega, \mathcal{F}, P)$ some abstract probability space with P a probability measure on $\mathcal{F}$ a σ -field of events in Ω . The expectation of a random variable on $(\Omega, \mathcal{F}, P)$ will be denoted by $\mathbb{E}$.

The attitude of a decision maker towards risk will involve a utility function. Throughout this paper we shall deal with the following classes of utility functions.

Definition 2.1 Let $u : \mathbb{R} \rightarrow [-\infty, +\infty)$ be a proper closed concave and nondecreasing utility function with effective domain $\text{dom } u = \{t \in \mathbb{R} \mid u(t) > -\infty\} \neq \emptyset$. Assume that u satisfies

$$u(0) = 0, \quad 1 \in \partial u(0), \quad (2.1)$$

where $\partial u(\cdot)$ denotes the subdifferential map of u . Such a class is denoted by $\mathbb{U}_0$.

Recall that a utility function is unique, up to a monotone increasing affine transformation. The conditions (2.1) are easy to enforce and can be seen as producing *normalized* utilities. For every $u \in \mathbb{U}_0$ one has:

$$u(x) \geq 0, \quad \forall x \geq 0 \text{ [since } u \text{ is nondecreasing]}, \quad (2.2)$$

$$u(x) \leq x, \quad \forall x \text{ [since } u \text{ is concave]}. \quad (2.3)$$

The main thesis underlying the OCE is that *assigning a value to a random variable is in itself a decision problem*.

Definition 2.2 Let X be a random variable and let $u \in \mathbb{U}_0$. The *Optimized Certainty Equivalent (OCE)* of the uncertain outcome X is defined by

$$S_u(X) = \sup_{\eta \in \mathbb{R}} \{\eta + \mathbb{E}u(X - \eta)\}. \quad (2.4)$$

To analyze the properties of $S_u(\cdot)$, we first need to recall some basic facts and properties of concave functions of one variable, see e.g., [26, Sections 23, 24] for details and proofs.

Lemma 2.1 Let $u : \mathbb{R} \rightarrow [-\infty, +\infty)$ be proper, closed and concave. Then, the right derivative u'_+ and the left derivative u'_- exists as extended real numbers, and the following properties hold:

(a) u'_+, u'_- are nonincreasing on $\mathbb{R}$, finite on the interior of $\text{dom } u$, ($\text{int dom } u$), and such that:

$$u'_+(a) \geq u'_-(t) \geq u'_+(t) \geq u'_-(b), \quad \forall a < t < b. \quad (2.5)$$

(b) If $\text{dom } u$ is a closed bounded interval $[a, b]$, then $u'_+(a)$ exists but may be equal to $+\infty$, and $u'_-(b)$ exists but may be equal to $-\infty$.

(c) The subdifferential of u is given by

$$\partial u(t) = \{s \in \mathbb{R} \mid u'_+(t) \leq s \leq u'_-(t)\}, \quad \forall t. \quad (2.6)$$

Moreover, if $\text{dom } u = [a, b]$, a closed bounded interval, then $\partial u(t) = \emptyset$ for $t < a$ or $t > b$ and

$$\partial u(t) = \begin{cases} (-\infty, u'_-(a)] & \text{for } t = a, \\ [u'_+(t), u'_-(t)] & \text{for } a < t < b, \\ (+\infty, u'_+(b)] & \text{for } t = b. \end{cases} \quad (2.7)$$

In the sequel, we freely interchange integration (through the expectation operator) with one sided derivation, and thus implicitly assume that these operations are well defined, (this is the case for example under the assumption on the continuity of the one sided derivatives of u and the finiteness of associated expectations, see e.g., [30]).

The following proposition states that whenever there is an optimal solution to the problem (2.4), there is an optimal solution in the support of the random variable X , (denoted by $\text{supp}X$).

Proposition 2.1 *Let $u \in \mathbb{U}_0$, and let X be a random variable with support $\text{supp} X = [x_{\min}, x_{\max}]$, ($-\infty \leq x_{\min} < x_{\max} \leq +\infty$), such that*

$$\mathcal{E}^* := \{\eta^* \in \mathbb{R} \mid S_u(X) = \eta^* + \mathbb{E}u(X - \eta^*)\} \neq \emptyset.$$

Then, $\text{supp} X \cap \mathcal{E}^ \neq \emptyset$, and one has*

$$S_u(X) = \sup_{\eta \in [x_{\min}, x_{\max}]} \{\eta + \mathbb{E}u(X - \eta)\}. \quad (2.8)$$

In particular, when $\text{supp} X$ is a closed bounded interval, the supremum is attained.

Proof. First note that for any $u \in \mathbb{U}_0$, one has $1 \in \partial u(0)$, and hence, by Lemma 2.1(c), that $u'_+(0) \leq 1 \leq u'_-(0)$. Now, let $g(\eta) := \eta + \mathbb{E}u(X - \eta)$. Then, $g(\cdot)$ is a closed proper concave function. Writing the necessary and sufficient optimality conditions for the concave problem $\sup\{g(\eta) \mid \eta \in \mathbb{R}\}$, one has $\eta^* \in \arg\max g(\eta) \equiv \mathcal{E}^*$ if and only if η^* satisfies $0 \in \partial g(\eta^*)$, which by Lemma 2.1(c) reduces to

$$\mathbb{E}u'_+(X - \eta^*) \leq 1 \leq \mathbb{E}u'_-(X - \eta^*). \quad (2.9)$$

Suppose $\eta^* \notin [x_{\min}, x_{\max}]$, then $\eta^* > x_{\max}$, or $\eta^* < x_{\min}$. If $\eta^* > x_{\max}$, then $X - x_{\max} > X - \eta^*$, and hence by the monotonicity of u'_- (see, Lemma 2.1(a)), it follows that $\mathbb{E}u'_-(X - x_{\max}) < \mathbb{E}u'_-(X - \eta^*) \leq 1$. But, $X - x_{\max} \leq 0$ with probability 1, thus by the monotonicity of u'_- and since $u'_-(0) \geq 1$, one obtains $\mathbb{E}u'_-(X - x_{\max}) \geq u'_-(0) \geq 1$, and thus it follows that $\mathbb{E}u'_-(X - x_{\max}) = 1$, proving that $x_{\max} \in \mathcal{E}^*$. Similarly, if $\eta^* < x_{\min}$, one obtains $x_{\min} \in \mathcal{E}^*$, and thus the claimed statement (2.8) follows, with attainment of the supremum, whenever the support of X is assumed to be a closed and bounded interval. $\square$

Remark 2.1 Proposition 2.1 was proved in [6] under the sharper assumption that u is a continuously differentiable and strictly concave function. Note that in this case, and with $\text{supp}X$ assumed to be a compact support, the supremum in (2.8) is uniquely attained at $\eta_s \in [x_{\min}, x_{\max}]$ which is solution of

$$\mathbb{E}u'(X - \eta_s) = 1, \quad (2.10)$$

and one has

$$S_u(X) = \eta_s + \mathbb{E}u(X - \eta_s). \quad (2.11)$$

The appropriateness of S_u as a certainty equivalent measure is supported by its fundamental properties given below, that were originally proved in our work [6]. To make the paper self contained, we include the proof.

Theorem 2.1 (*Main Properties of OCE*) For any utility $u \in \mathbb{U}_0$, and any random variable X the following properties hold:

- (a) **Shift Additivity.** $S_u(X + c) = S_u(X) + c, \forall c \in \mathbb{R}$.
- (b) **Consistency.** $S_u(c) = c$, for any constant $c \in \mathbb{R}$ (considered as a degenerate random variable).
- (c) **Monotonicity.** Let Y be any random variable such that $X(\omega) \leq Y(\omega), \forall \omega \in \Omega$. Then, $S_u(X) \leq S_u(Y)$.
- (d) **Concavity.** For any random variables X_1, X_2 and any $\lambda \in (0, 1)$, one has

$$S_u(\lambda X_1 + (1 - \lambda)X_2) \geq \lambda S_u(X_1) + (1 - \lambda)S_u(X_2).$$

Proof. (a) For any $u : \mathbb{R} \rightarrow [-\infty, +\infty)$, and any $c \in \mathbb{R}$,

$$\begin{aligned} S_u(X + c) &= \sup_{\eta \in \mathbb{R}} \{\eta + \mathbb{E}u(X + c - \eta)\} \\ &= c + \sup_{\eta \in \mathbb{R}} \{\eta - c + \mathbb{E}u(X - (\eta - c))\} = c + S_u(X). \end{aligned}$$

(b) Since $u \in \mathbb{U}_0$, then $u(0) = 0$, $1 \in \partial u(0)$ and the concavity of u implies $u(t) \leq t$, and hence

$$S_u(X) \leq \sup_{\eta \in \mathbb{R}} \{\eta + (c - \eta)\} = c.$$

For the converse inequality, since $u(0) = 0$, one has $S_u(c) \geq \{c + u(c - c)\} = c$.

(c) If $X \leq Y$, then $X - \eta \leq Y - \eta$, and since u is nondecreasing it follows that,

$$S_u(X) = \sup_{\eta \in \mathbb{R}} \{\eta + \mathbb{E}u(X - \eta)\} \leq \sup_{\eta \in \mathbb{R}} \{\eta + \mathbb{E}u(Y - \eta)\} = S_u(Y).$$

(d) Let $\lambda \in (0, 1)$ and for any random variables X_1, X_2 , let $X_\lambda := \lambda X_1 + (1 - \lambda)X_2$. Since u is concave, the function $f(z, \eta) := \eta + u(z - \eta)$ is jointly concave over $\mathbb{R} \times \mathbb{R}$. Therefore, for any $\eta_1, \eta_2 \in \mathbb{R}$, and with $\eta_\lambda := \lambda \eta_1 + (1 - \lambda)\eta_2$, one has,

$$\mathbb{E}f(X_\lambda, \eta_\lambda) \geq \lambda \mathbb{E}f(X_1, \eta_1) + (1 - \lambda)\mathbb{E}f(X_2, \eta_2).$$

Since $S_u(X_\lambda) = \sup_{\eta \in \mathbb{R}} \mathbb{E}f(X_\lambda, \eta)$, it follows that,

$$S_u(X_\lambda) \geq \sup_{\eta_1, \eta_2} \{\lambda \mathbb{E}f(X_1, \eta_1) + (1 - \lambda)\mathbb{E}f(X_2, \eta_2)\} = \lambda S_u(X_1) + (1 - \lambda)S_u(X_2).$$

□

Remark 2.2 (i) Note that for the consistency property it is enough to assume that $u(t) \leq t, \forall t$.
(ii) The shift-additivity of $S_u(\cdot)$ is a *generic property* of the OCE, and holds for any function u and any random variable X . Recall that two other certainty equivalents for which the shift additivity property holds are the so called u-mean (see Buhlman [12]) and the Yaari's certainty equivalent, [31]. On the other hand, for the classical certainty equivalent $C_u(X) := u^{-1}\mathbb{E}u(X)$, the shift additivity property holds if and only if u is a linear or exponential utility function, see Section 5

Further important properties and examples of the OCE will be given below. At this point, we are ready to show that the OCE provides the basis for a natural measure of risk, that will be shown to encompass several important risk measures recently proposed in the literature.

2.2 OCE and Risk Measures: The missing link

In a fundamental paper[2], Artzner et al. have introduced the notion of a coherent risk measure within an axiomatic framework, as defined below. Let $\mathcal{X}$ be a linear space of functions on Ω to the real line.

Definition 2.3 *A map $\rho : \mathcal{X} \rightarrow \mathbb{R}$ is called a coherent risk measure if and only if the following properties hold:*

- (a) *Subadditivity* : $\rho(X + Y) \leq \rho(X) + \rho(Y)$.
- (b) *Positive homogeneity*: If $\lambda \geq 0$, then $\rho(\lambda X) = \lambda\rho(X)$.
- (c) *Monotonicity*: If $X \leq Y$ (i.e. $X(\omega) \leq Y(\omega)$, $\forall \omega \in \Omega$) then $\rho(X) \geq \rho(Y)$.
- (d) *Translation invariance*: If $m \in \mathbb{R}$, then $\rho(X + m) = \rho(X) - m$.

More recently, Föllmer and Schied [15], and Frittelli and Gianin [14] proposed to extend the concept coherent risk measure by relaxing the subadditivity and positive homogeneity conditions¹ with the weaker convexity requirement:

$$(e) \text{ Convexity: } \rho(\lambda X + (1 - \lambda)Y) \leq \lambda\rho(X) + (1 - \lambda)\rho(Y), \forall \lambda \in [0, 1].$$

Definition 2.4 *A map $\rho : \mathcal{X} \rightarrow \mathbb{R}$ is called a convex risk measure if and only if satisfies the condition of monotonicity (c), translation invariance (d), and convexity (e).*

The axiomatic framework of convex risk measures given in Definition 2.4, together the properties of the OCE given in Theorem 2.1 immediately implies that the *negative* of the OCE satisfies the properties (c), (d) and (e) of Definition 2.4, i.e.,

$$\rho(X) := -S_u(X),$$

is a convex risk measure. Thus, the OCE provides a simple way to generate convex risk measures via particular choices of a utility function. We now illustrate this fact by some interesting examples, more will be given in Sections 4, 4.3 and 5.

2.3 Examples of Convex Risk Measures via OCE

We compute the OCE for several utility functions. The corresponding risk measures, as explained, are just the negative of the OCE.

Example 2.1 (*Exponential Utility Function*) Let $u(t) = 1 - e^{-t}$, $\forall t \in \mathbb{R}$. Using, (2.10), one obtains $\mathbb{E}(e^{-X+\eta_s}) = 1$, giving $\eta_s = -\log \mathbb{E}e^{-X}$, which in turns is also the value of OCE, i.e.,

$$S_u(X) = -\log \mathbb{E}e^{-X} (= \eta_s).$$

It is interesting to note that for this exponential utility *all* the three risk measures C_u , M_u , and S_u coincide.

¹Recall that if a function f is positively homogeneous and subadditive, then f is convex, [26].

Example 2.2 (*Quadratic Utility*) Let

$$u(t) = \begin{cases} t - \frac{1}{2}t^2 & \text{if } t \leq 1, \\ -\infty & \text{otherwise,} \end{cases}$$

and let X be a random variable with right support $x_{\max} \leq 1$. Denote $\mu := \mathbb{E}(X)$ and σ^2 the variance of X , which are assumed to exist. A direct computation from (2.10) yields,

$$S_u(X) = \mu - \frac{1}{2}\sigma^2.$$

The next example shows that the popular risk measure CVAR (conditional value at risk) described by its minimization formula as given by Rockafellar and Uryasev [27], is just a special case of the OCE with the choice of a piecewise linear utility.

Example 2.3 (*Piecewise linear utility function*) Let

$$u(t) = \begin{cases} \gamma_2 t & \text{if } t \leq 0, \\ \gamma_1 t & \text{if } t > 0, \end{cases}$$

where $0 \leq \gamma_1 < 1 < \gamma_2$. Note that the latter choice for (γ_1, γ_2) guarantees that $1 \in \partial u(0)$ and hence $u \in \mathbb{U}_0$. For any real number z , denote $[z]_+ = \max\{0, z\}$. Then, the utility function can be written as

$$u(t) = \gamma_1[t]_+ - \gamma_2[-t]_+,$$

and the OCE yields,

$$S_u(X) = \sup_{\eta \in \mathbb{R}} \{\eta - \gamma_2 \mathbb{E}[\eta - X]_+ + \gamma_1 \mathbb{E}[X - \eta]_+\}. \quad (2.12)$$

Thus, with $\rho(X) := -S_u(X)$, we obtain the convex and coherent risk measure

$$\rho(X) = \inf_{\eta \in \mathbb{R}} \{\eta - \gamma_2 \mathbb{E}(-\eta - X)_+ - \gamma_1 \mathbb{E}(X + \eta)_+\},$$

which recovers the recent related risk measure considered in [24].

With $\gamma_1 = 0$, and $\alpha := 1/\gamma_2 \in (0, 1)$ (recall that here we assumed $\gamma_2 > 1$), we obtain as a special case of the last equation, the representation formula for the conditional value at risk CVAR_α , (see [27]), i.e.,

$$\text{CVAR}_\alpha(X) = \inf_{\eta \in \mathbb{R}} \left\{ \eta + \frac{1}{\alpha} \mathbb{E}(X + \eta)_- \right\},$$

where for any $z \in \mathbb{R}$, we denote $[z]_- = [-z]_+$.

The optimization problem (2.12) can be solved analytically. Let F be the distribution function of the random variable X , i.e., $F(x) = P\{X \leq x\}$, and let $F^{-1}(y) := \inf\{x \mid F(x) > y\}$ be its right continuous inverse. For a confidence level $\alpha \in (0, 1)$, $F^{-1}(\alpha)$ is the α -quantile of the random variable X (also called the *the value at risk* and often denoted by VaR_α). In that case, the optimal solution η^* maximizing (2.12) is given by

$$\eta^* = F^{-1} \left(\frac{1 - \gamma_1}{\gamma_2 - \gamma_1} \right),$$

and the associated OCE is

$$S_u(X) = \gamma_2 \int_{-\infty}^{\eta^*} t dF(t) + \gamma_1 \int_{\eta^*}^{+\infty} t dF(t).$$

Furthermore, for the special case $\gamma_1 = 0$, and $\alpha := 1/\gamma_2 \in (0, 1)$, (i.e., $u(t) = \min\{0, \alpha^{-1}t\}$) the latter formula gives

$$S_u(X) = \frac{1}{\alpha} \int_{-\infty}^{F^{-1}(\alpha)} t dF(t) = \frac{1}{\alpha} \int_0^\alpha F^{-1}(v) dv = \frac{1}{\alpha} \int_0^\alpha \text{VaR}_v dv = \text{CVAR}_\alpha(X),$$

recovering the integral formula of Acerbi [1] for CVAR.

Some other derivations of CVAR will be further discussed in Section 4.3.

2.4 Homogeneity, Ranking and Limiting Properties of OCE

Decision making under uncertainty assumes the ability of ranking random variables. Expected utility theory is the classical theory for decision making under risk. Under the axiomatic foundation of von Neumann and Morgenstern, [22] a decision maker is assumed to have a utility function, and a random variable X is preferred to Y , which we denote $X \succeq Y$, if and only if $\mathbb{E}u(X) \geq \mathbb{E}u(Y)$. A decision maker is *risk averse* if for any random variable X , its expectation $\mathbb{E}(X)$ is preferred to X , namely

$$u(\mathbb{E}(X)) \geq \mathbb{E}u(X),$$

which in turn is just the Jensen inequality for u , i.e., is equivalent to say that the utility u is concave. If u is also assumed strictly increasing, then the later inequality translates risk aversion to,

$$\mathbb{E}(X) \geq u^{-1}\mathbb{E}u(X) \equiv C_u(X), \text{ for any random variable } X,$$

where $C_u(\cdot)$ is the classical certainty equivalent.

Like the Expected Utility model, the OCE also exhibits the risk aversion property, but under weaker assumptions on the utility than concavity and monotonicity.

Proposition 2.2 (*Risk Aversion*) *Let $u : \mathbb{R} \rightarrow [-\infty, +\infty)$ be a closed proper function. Then,*

$$S_u(X) \leq \mathbb{E}(X), \text{ for any random variable } X, \text{ if and only if } u(t) \leq t, \forall t \in \mathbb{R}.$$

Proof. Suppose that for any random variable X one has $S_u(X) \leq \mathbb{E}(X)$. Then, for any random variable X and any $\eta \in \mathbb{R}$, $\eta + \mathbb{E}u(X - \eta) \leq \mathbb{E}(X)$, and hence, $\mathbb{E}u(X - \eta) \leq \mathbb{E}(X - \eta)$, so that $\mathbb{E}u(X) \leq \mathbb{E}(X)$, which proves $u(t) \leq t$. Conversely, if the later inequality holds, then for any random variable X , $S_u(X) = \sup_{\eta \in \mathbb{R}} \{\eta + \mathbb{E}u(X - \eta)\} \leq \sup_{\eta \in \mathbb{R}} \{\eta + \mathbb{E}(X - \eta)\} = \mathbb{E}(X)$. $\square$

Let $u \in \mathbb{U}_0$. For any $\delta > 0$, consider a one parameterized utility,

$$u_\delta(t) := \frac{u(\delta t)}{\delta}. \tag{2.13}$$

The parameter δ can be seen to measure the degree of risk aversion, (see next Section for details.) As an example, if $u(t) = 1 - e^{-t}$, then by Example (2.1) one obtains,

$$S_{u_\delta}(X) = -\delta^{-1} \log \mathbb{E}(e^{-\delta X}).$$

As the parameter δ increases from 0 to ∞ the degree of risk aversion increases. The limiting cases, $\delta = 0$ and $\delta = \infty$ correspond to risk neutrality (the certainty equivalent $C_0(X) = E(X)$ for the former and extreme risk aversion ($C_\infty(X) = x_{\min}$) for the latter, [4, Theorems 1 and 2,]; these results are a special case of Theorem 2.2 given below.

Consider an arbitrary but fixed utility $u \in \mathbb{U}_0$. For any $\delta > 0$, let u_δ be as defined in (2.13). The corresponding OCE is:

$$\begin{aligned} S_\delta(X) := S_{u_\delta}(X) &= \sup_{\eta} \left\{ \eta + \frac{1}{\delta} \mathbb{E} u(\delta(X - \eta)) \right\} \\ &= \frac{1}{\delta} \sup_{\eta} \left\{ \eta + \mathbb{E} u(\delta X - \eta) \right\} \\ &= \frac{1}{\delta} S_u(\delta X). \end{aligned} \tag{2.14}$$

The next result shows that $S_\delta(X)$ is a nonincreasing function of δ , which means that the OCE possesses the subhomogeneity property. This property is one which distinguishes convex risk measures from coherent ones, the latter requiring homogeneity.

Proposition 2.3 (*Subhomogeneity*) *For any $u \in \mathbb{U}_0$ and for any random variable X , the optimized certainty equivalent $S_u(X)$ is subhomogeneous, i.e.,*

$$\begin{aligned} (a) \quad & S_u(\delta X) \leq \delta S_u(X), \quad \forall \delta > 1, \\ (b) \quad & S_u(\delta X) \geq \delta S_u(X), \quad \forall 0 \leq \delta \leq 1. \end{aligned}$$

Proof. We will show that the function $s(\delta) := \frac{1}{\delta} S_u(\delta X)$ is nonincreasing in $\delta > 0$ for every random variable X , which clearly implies (a) and (b). Now, by (2.14) one has,

$$s(\delta) = \frac{1}{\delta} S_u(\delta X) = \sup_{\eta} \left\{ \eta + \mathbb{E} \frac{1}{\delta} u(\delta(X - \eta)) \right\}. \tag{2.15}$$

Let $\delta_2 > \delta_1 > 0$. From (2.15) it follows that for any random variable X , one has $s(\delta_2) \leq s(\delta_1)$ if

$$\frac{1}{\delta_2} u(\delta_2 t) \leq \frac{1}{\delta_1} u(\delta_1 t), \quad \forall t \in \mathbb{R}. \tag{2.16}$$

Now, by the concavity of u and $\delta_2 > \delta_1 > 0$:

$$\frac{u(\delta_2 t) - u(\delta_1 t)}{\delta_2 - \delta_1} \leq \frac{u(\delta_1 t) - u(0)}{\delta_1 - 0}. \tag{2.17}$$

Since $u(0) = 0$, inequality (2.17) reduces to (2.16) and the proof is complete. $\square$

The next result gives limiting properties of the OCE.

Theorem 2.2 *Let u be an arbitrary but fixed utility function in $\mathbb{U}_0$, and let $u_\delta(t) := \frac{u(\delta t)}{\delta}$. Then,*
 (a) *For any random variable X with lower support $x_{\min} > -\infty$:*

$$\lim_{\delta \rightarrow \infty} S_\delta(X) = x_{\min}.$$

(b) *Moreover, if u is continuously differentiable on $\text{int dom } u$, then*

$$\lim_{\delta \rightarrow 0} S_\delta(X) = E(X), \quad (\text{Risk neutrality}).$$

Proof. For all $\delta > 0$, one has $\delta X - \eta \geq \delta x_{\min} - \eta$, and since u is monotone nondecreasing, it follows that

$$S_\delta(X) \geq \frac{1}{\delta} \sup_{\eta \in \mathbb{R}} \{\eta + u(\delta x_{\min} - \eta)\} = \frac{S_u(\delta x_{\min})}{\delta} = x_{\min},$$

where the last equality is by Theorem 2.1(b). The reverse inequality, needed to complete the proof of (a) is postponed until Section 4. The result given in (b) was proved in [7, Lemma 4.2]. $\square$

Ranking with OCE

The OCE may induce a different order on random variables than the one induced by the classical certainty equivalent $C_u(X) = u^{-1}Eu(X)$, i.e., for *some* $u \in \mathbb{U}_0$, and some random variables X, Y :

$$C_u(X) \geq C_u(Y) \tag{2.18}$$

but

$$S_u(X) < S_u(Y).$$

However, if (2.18) holds for *all* $u \in \mathbb{U}_0$, then also $S_u(X) \geq S_u(Y)$ for all $u \in \mathbb{U}_0$. This means that the ranking of random variables imposed by the OCE is consistent with the ranking imposed by *2nd order stochastic dominance*, (see [18]). The following result was proven in [5, Theorem 2.1(g)], and for completeness we include its simple proof.

Proposition 2.4 (*2nd order stochastic dominance*) *Let X, Y be random variables with compact support. Then, for all $u \in \mathbb{U}_0$,*

$$S_u(X) \geq S_u(Y) \iff C_u(X) \geq C_u(Y).$$

Proof. Since u is nondecreasing, then assuming that $Eu(X) \geq Eu(Y)$, and using the definition of S_u it follows that $S_u(X) \geq S_u(Y)$. To prove the reverse implication, invoking Proposition 2.1 with η_X, η_Y being the points where the supremum defining $S_u(X)$ and $S_u(Y)$ are respectively attained, one obtains for any $u \in \mathbb{U}_0$

$$\begin{aligned} S_u(X) &= \eta_X + Eu(X - \eta_X) \geq \eta_Y + Eu(Y - \eta_Y) \quad (\text{since } S_u(X) \geq S_u(Y)) \\ &\geq \eta_X + Eu(Y - \eta_X). \end{aligned}$$

Therefore, for any $u \in \mathbb{U}_0$, $Eu(X - \eta_X) \geq Eu(Y - \eta_X)$, which implies $Eu(X) \geq Eu(Y)$. $\square$

2.5 Recovering u from its OCE

Thus far we have studied the properties of S_u induced by its utility function u . Here, we consider the inverse problem, namely of recovering the utility function u from a given OCE. For that purpose, we introduce the class $\mathbb{U}_0^<$ of *strongly risk averse utilities*, i.e., all $u \in \mathbb{U}_0$ such that

$$u(t) < t \quad \forall t \neq 0. \quad (2.19)$$

(Recall that a weak inequality holds for all $u \in \mathbb{U}_0$).

Let X_p be the random variable:

$$X_p = \begin{cases} x & \text{in prob. } p \\ 0 & \text{in prob. } 1 - p \end{cases} \quad (0 < p < 1, x > 0).$$

For a given $u \in \mathbb{U}_0$, the OCE of X_p will be denoted by $S_u[x, p]$, i.e.,

$$S_u[x, p] = S_u(X_p) = \sup_{0 \leq \eta \leq x} \{ \eta + pu(x - \eta) + (1 - p)u(-\eta) \}. \quad (2.20)$$

The following result shows how to extract the utility function $u(\cdot)$ from knowledge of the OCE $S_u[x, p]$.

Proposition 2.5 *If $u \in \mathbb{U}_0^<$ then*

$$\lim_{p \rightarrow 0^+} \frac{S_u[x, p]}{p} = u(x). \quad (2.21)$$

Proof. The optimal solution in the right-hand side of (2.20) is $\eta^* = 0$ if and only if (recall $u(0) = 0$)

$$\forall 0 < \eta \leq x : \quad \eta + pu(x - \eta) + (1 - p)u(-\eta) \leq pu(x),$$

or equivalently

$$\forall 0 < \eta \leq x : \quad p[u(x - \eta) - u(-\eta) - u(x)] \leq -u(-\eta) - \eta. \quad (2.22)$$

Now, $\eta > 0$ implies $u(-\eta) < -\eta$, so the right-hand side of (2.22) is strictly positive, and therefore for $p > 0$ sufficiently small, $\eta^* = 0$, is indeed the optimal solution, i.e., (2.22) holds. Summarizing: $\exists \bar{p} > 0$ such that $S_u[x, p] = pu(x)$, $\forall 0 < p < \bar{p}$, and consequently (2.21) follows. $\square$

Example 2.4 Let $S_u(X) = -\log(\mathbb{E}e^{-X})$. To recover u we first note that in that case,

$$S_u[x, p] = -\log(pe^{-x} + (1 - p)).$$

Then,

$$u(x) = \lim_{p \rightarrow 0^+} \frac{S_u[x, p]}{p} = \lim_{p \rightarrow 0^+} \frac{-\log(pe^{-x} + 1 - p)}{p} = \lim_{p \rightarrow 0^+} \frac{-(e^{-x} - 1)}{pe^{-x} + 1 - p} \quad (\text{by l'hospital rule}),$$

and hence, the recovered utility function is $u(x) = 1 - e^{-x}$.

Remark 2.3 The assumption of strong risk averse utilities in Proposition 2.5 is essential. A counterexample is offered by

$$\bar{u}(x) = \begin{cases} 1 - e^{-x} & \text{if } x \geq 0 \\ x & \text{if } x \leq 0. \end{cases}$$

Here a simple calculation shows that $S_{\bar{u}}[x, p] = px$, and hence

$$\lim_{p \rightarrow 0^+} \frac{S_{\bar{u}}[p, x]}{p} = x \neq \bar{u}(x).$$

3 When is the OCE a coherent risk measure?

In this section we characterize the subclass of utilities $u \in \mathbb{U}_0^<$ such that $\rho(X) = -S_u(X)$ is a coherent risk measure.

Theorem 3.1 In the class, $\mathbb{U}_0^<$ of strongly risk averse utilities, $\rho(X) = -S_u(X)$ is a coherent risk measure if and only if u is the piecewise linear function given by (cf. Example 2.3):

$$u(t) = \begin{cases} \gamma_2 t & \text{if } t \leq 0 \\ \gamma_1 t & \text{if } t > 0, \end{cases}$$

for some $\gamma_2 > 1 > \gamma_1 \geq 0$. □

The proof of Theorem 3.1 is an immediate corollary of the next two results.

Proposition 3.1 Let $u \in \mathbb{U}_0^<$. Then $S_u(X)$ is positively homogeneous for all random variables X if and only if u is positively homogeneous.

Proof. Let $u \in \mathbb{U}_0^<$. The “if” part of the statement is straightforward from the definition of $S_u(X)$. To prove the necessary condition, for $\alpha > 0 > \beta$, we consider the random variable X with $\Pr(X = \alpha) = p$, $\Pr(X = \beta) = 1 - p$, and the corresponding OCE,

$$S_u(X) = \sup_{\eta} \{ \eta + pu(\alpha - \eta) + (1 - p)u(\beta - \eta) \}. \quad (3.23)$$

For any $u \in \mathbb{U}_0^<$ we have (see Lemma 2.1)

$$\begin{cases} \forall \alpha > 0 > \beta : \\ u'_-(\alpha) \geq u'_+(\alpha) > u'_-(0) \geq 1 \geq u'_+(0) > u'_-(\beta) \geq u'_+(\beta) > 0. \end{cases} \quad (3.24)$$

Now, Let η^* be an optimal solution for (3.23), then it satisfies the following optimality conditions:

$$pu'_-(\alpha - \eta^*) + (1 - p)u'_-(\beta - \eta^*) \geq 1 \geq pu'_+(\alpha - \eta^*) + (1 - p)u'_+(\beta - \eta^*) .$$

In particular $\eta^* = 0$ if and only if

$$pu'_-(\alpha) + (1 - p)u'_-(\beta) \geq 1 \geq pu'_+(\alpha) + (1 - p)u'_+(\beta) \quad (3.25)$$

which holds if and only if

$$\frac{1 - u'_-(\beta)}{u'_-(\alpha) - u'_-(\beta)} \leq p \leq \frac{1 - u'_+(\beta)}{u'_+(\alpha) - u'_+(\beta)}. \quad (3.26)$$

Note that $p \in (0, 1)$ by (3.24). Moreover, to be well defined, the left hand side of (3.26) must be less than or equal to the right hand side or:

$$\underbrace{(1 - u'_-(\beta))(u'_+(\alpha) - u'_+(\beta))}_A \leq \underbrace{(1 - u'_+(\beta))(u'_-(\alpha) - u'_-(\beta))}_B.$$

Now,

$$\begin{aligned} B - A &= u'_-(\alpha)(1 - u'_+(\beta)) - u'_-(\beta) - u'_+(\alpha) + u'_+(\alpha)u'_-(\beta) + u'_+(\beta) \\ \text{[by (3.24)]} &\geq u'_+(\alpha)(1 - u'_+(\beta) - u'_-(\beta) - u'_+(\alpha) + u'_+(\alpha)u'_-(\beta) + u'_+(\beta)) \\ &= u'_+(\alpha)[1 - u'_+(\beta) - 1 + u'_-(\beta)] - u'_-(\beta) + u'_+(\beta) \\ &= [u'_-(\beta) - u'_+(\beta)][u'_+(\alpha) - 1] \geq 0 \end{aligned}$$

the last inequality again by (3.24).

Let then $p_0 \in (0, 1)$ be a probability satisfying (3.26) and let X_0 be the random variable:

$$\Pr(X_0 = \alpha) = p_0, \quad \Pr(X_0 = \beta) = 1 - p_0$$

where $\alpha > 0 > \beta$. For this random variable (recall $\eta^* = 0$)

$$S_u(X_0) = p_0 u(\alpha) + (1 - p_0)u(\beta).$$

Let $0 < \lambda < 1$. Then (since $\eta^* = 0$ is not necessarily the optimal solution for the random variable λX_0),

$$\begin{aligned} S_u(\lambda X_0) &\geq p_0 u(\lambda \alpha) + (1 - p_0)u(\lambda \beta) \\ &= p_0 u(\lambda \alpha + (1 - \lambda)0) + (1 - p_0)(u(\lambda \beta + (1 - \lambda)0)) \\ &\geq p_0 \lambda u(\alpha) + (1 - p_0) \lambda u(\beta) \text{ [by concavity of } u \text{ and } u(0) = 0] \\ &= \lambda [p_0 u(\alpha) + (1 - p_0)u(\beta)] = \lambda S_u(X_0). \end{aligned}$$

But, by assumption, $S_u(\lambda X_0) = \lambda S_u(X_0)$, hence *equalities* must hold in the above chain of inequalities. In particular then

$$p_0 u(\lambda \alpha) + (1 - p_0)u(\lambda \beta) = p_0 \lambda u(\alpha) + (1 - p_0) \lambda u(\beta),$$

or equivalently,

$$p_0 [u(\lambda \alpha) - \lambda u(\alpha)] + (1 - p_0) [u(\lambda \beta) - \lambda u(\beta)] = 0. \quad (3.27)$$

Each of the square brackets in equation (3.27) is nonnegative, again by the concavity of u ; hence we proved

$$\forall 0 < \lambda < 1 : \begin{cases} u(\lambda \alpha) = \lambda u(\alpha) & \forall \alpha > 0 \\ u(\lambda \beta) = \lambda u(\beta) & \forall \beta < 0 \end{cases}$$

and since $u(0) = 0$ we have:

$$\forall 0 < \lambda < 1 : u(\lambda x) = \lambda u(x), \quad \forall x. \quad (3.28)$$

Now, let $\lambda > 1$, then by (3.28): $u(\lambda^{-1}x) = \lambda^{-1} u(x)$, $\forall x$. In particular, for x replaced by λx we thus have for all $\lambda > 1$: $u(x) = \lambda^{-1}u(\lambda x)$, $\forall x$. $\square$

Next, we characterize utilities $u \in \mathbb{U}_0^<$ which are positively homogeneous.

Lemma 3.1 *Let $u \in \mathbb{U}_0^<$ be a finite positively homogeneous utility function. Then u is a piecewise linear function of the form*

$$u(x) = \begin{cases} \gamma_2 x & x \leq 0 \\ \gamma_1 x & x > 0 \end{cases} \quad (3.29)$$

where $\gamma_2 > 1 > \gamma_1 \geq 0$.

Proof Let $f(x) = -u(-x)$, then f is finite convex positively homogeneous function, and hence it is the support function of a bounded convex set, i.e., a closed interval $[\gamma_1, \gamma_2]$ (see [26, Cor. 13.2.2]). Hence, $f(x)$ can be represented as:

$$f(x) = -u(-x) = \sup_{\gamma_1 \leq y \leq \gamma_2} y^T x \quad \text{for some } \gamma_2 \geq \gamma_1,$$

and hence,

$$-u(-x) = \begin{cases} \gamma_1 x & \text{if } x \leq 0 \\ \gamma_2 x & \text{if } x \geq 0, \end{cases}$$

which gives (3.29). Since $u \in \mathbb{U}_0^<$, for $x \neq 0$, one has $u(x) < x$, so we must have $\gamma_2 > 1 > \gamma_1$, and since u is nondecreasing also that $\gamma_1 \geq 0$. $\square$

Remark 3.1 The assumption $u \in \mathbb{U}_0^<$ (and not just $u \in \mathbb{U}_0$) is crucial for the validity of the coherence result in Theorem 3.1. Without this assumption, there exist examples of *nonhomogeneous* $u \in \mathbb{U}_0$ for which $S_n(X)$ is homogeneous (coherent) for some random variable X . One such example is the following

$$\bar{u}(t) = \begin{cases} 1 - e^{-t} & t \geq 0 \\ t & t \leq 0. \end{cases}$$

Then $\bar{u} \in \mathbb{U}_0$ but $\bar{u}(t) \not\leq t$ for $t < 0$, hence, $\bar{u} \notin \mathbb{U}_0^<$. Let X be a random variable, and denote its right support by $x_{\max}$, i.e.,

$$\Pr\{X \leq x_{\max}\} = 1.$$

Then, writing $S_u(\cdot)$ in terms of conditional expectations one has,

$$\begin{aligned} S_{\bar{u}}(X) &= \sup_{\eta \leq x_{\max}} \{\eta + \mathbb{E}\bar{u}(X - \eta)\} \\ &= \sup_{\eta \leq x_{\max}} \{\eta + \mathbb{E}(\bar{u}(X - \eta) \mid X > \eta) \Pr(X > \eta) \\ &\quad + \mathbb{E}(\bar{u}(X - \eta) \mid X \leq \eta) \Pr(X \leq \eta)\} \end{aligned}$$

$$\begin{aligned}
&= \sup\{\eta + \mathbb{E}((1 - e^{-(X-\eta)} \mid X > \eta) \Pr(X > \eta) \\
&\quad + \mathbb{E}(X - \eta \mid X \leq \eta)) \Pr(X \leq \eta)\}, \\
&\leq \sup_{\eta}\{\eta + \mathbb{E}(X - \eta)\} = \mathbb{E}(X), \text{ (since } 1 - e^{-t} \leq t\text{)}.
\end{aligned}$$

But this upper bound is attained by $\eta = x_{\max}$. Hence, we have proved that for any random variable X , $S_{\bar{u}}(X) = \mathbb{E}(X)$, which is certainly homogeneous, although $\bar{u}$ is *not* a homogeneous function.

4 Duality: Risk Measures and φ -Divergence

The OCE is directly related to the concept of φ -divergence which generalizes relative entropy measures and quantifies distance between random variables. In fact, the OCE originally emerged from using the concept of φ -entropic penalties we introduced in [7]. As shown there, convex duality theory plays the key role for deriving the OCE and providing a link between economics of uncertainty and information theory. In this section, we recall and extend some of the results we developed in [7] but within a more general setting. We also extend the extremal principles between certainty equivalents and generalized entropy we have established in [9] for discrete distributions, to the case of general distributions. Our results provide concrete realizations of the recent general representation theorems given in Föllmer and Shied [15] for general convex risk measures.

4.1 The φ -divergence functional

A well-known measure of distance between two random variables is the so-called Kullback-Liebler relative entropy distance, [19]. Here we consider a substantial generalization of relative entropy, the so-called φ -divergence introduced by Csiszár [13]. For a systematic treatment of φ -divergences, we refer the reader to the monograph of Liese and Vajda [20].

To proceed with the analysis we outline our setting and introduce some notations. Let $(\Omega, \mathcal{F})$ be a measurable space equipped with σ -algebra $\mathcal{F}$. Consider two probability distributions P and Q , and let μ be an arbitrary dominating positive measure of P and Q , (i.e., (Ω, μ) is a σ -finite measure space), such that both P and Q are absolutely continuous with respect to μ on $\mathcal{F}$, which we denote by $P \ll \mu$, $Q \ll \mu$. The Radon-Nikodym theorem implies that there exists an $\mathcal{F}$ -measurable function $x \geq 0$, ($y \geq 0$) the density of P , (Q) with respect to μ , and we write

$$x = \frac{dP}{d\mu}, \quad y = \frac{dQ}{d\mu}.$$

For $1 \leq p \leq +\infty$, let $L^p \equiv L^p(\Omega, \mathcal{F}, P)$ be the linear space of measurable real valued functions $f : \Omega \rightarrow \mathbb{R}$ with $\|f\|_p < \infty$, and where

$$\|f\|_p = \begin{cases} \left(\int_{\Omega} |f(\omega)|^p dP(\omega) \right)^{1/p} & \text{for } p \in [1, \infty) \\ \sup_{\omega \in \Omega} |f(\omega)| & \text{for } p = \infty. \end{cases}$$

For $p \in [1, +\infty)$, we denote by L^q its dual space with $q \in (1, +\infty]$, and $p + q = pq$. Furthermore, for $y \in L^p$ and $X \in L^q$, the scalar product

$$\langle y, X \rangle = \int_{\Omega} y(\omega) X(\omega) d\mu(\omega) = \int_{\Omega} X(\omega) \frac{dQ}{dP} dP(\omega) = \int_{\Omega} X(\omega) dQ(\omega) = \mathbb{E}_Q(X), \quad (4.30)$$

defines the pairing between L^p and L^q .

Let $\varphi : \mathbb{R} \rightarrow (-\infty, +\infty]$ be a proper closed convex function such that $\text{dom } \varphi$ is an interval with endpoints $\alpha < \beta$, so, $\text{int dom } \varphi = (\alpha, \beta)$. Since φ is closed, then

$$\lim_{t \rightarrow \alpha_+} \varphi(t) = \varphi(\alpha), \text{ if } \alpha \text{ is finite} \quad \lim_{t \rightarrow \beta_-} \varphi(t) = \varphi(\beta), \text{ if } \beta \text{ is finite.}$$

We assume that $1 \in \text{int dom } \varphi$ and that the minimum of φ is 0, and attained at the point $t^* = 1 \in \text{int dom } \varphi$.² The class of such function is denoted by Φ .

Definition 4.1 *Given $\varphi \in \Phi$, the φ -divergence of the probability measure Q with respect to P is*

$$I_\varphi(Q, P) = \begin{cases} \int_\Omega \varphi\left(\frac{dQ}{dP}\right) dP & \text{if } Q \ll P \\ +\infty & \text{otherwise.} \end{cases} \quad (4.31)$$

As an example, with $\varphi(t) = t \log t - t + 1$, the φ -divergence recovers the Kullback-Leibler relative entropy, and with $\varphi(t) = (\sqrt{t} - 1)^2$, one obtains the so-called Hellinger distance. For other many interesting choices of φ , see Csiszár [13] and the monograph of Liese and Vajda [20].

Thus, whenever $I_\varphi(Q, P)$ is finite, i.e., if $Q \ll P$, the φ -divergence can be conveniently written as

$$I_\varphi(Q, P) = \mathbb{E}_P \varphi\left(\frac{dQ}{dP}\right). \quad (4.32)$$

Now, since φ is convex, by Jensen's Inequality one has

$$I_\varphi(Q, P) \geq \varphi\left(\mathbb{E}_P\left(\frac{dQ}{dP}\right)\right) = \varphi(1) = 0,$$

with equality if $Q = P$ (since 1 is the point where φ attains its minimum 0), so that $I_\varphi(Q, P)$ is a measure of distance of Q from P .

Note that in terms of the densities x, y , the φ -divergence can also be written as, (see [13]),

$$I_\varphi(y, x) := \int_\Omega x(\omega) \varphi\left(\frac{y(\omega)}{x(\omega)}\right) d\mu(\omega),$$

where to avoid meaningless expressions, it is assumed that,

$$\varphi(0) < \infty, \quad 0\varphi\left(\frac{0}{0}\right) \equiv 0, \quad 0\varphi\left(\frac{s}{0}\right) = \lim_{\varepsilon \rightarrow 0} \varepsilon \varphi\left(\frac{s}{\varepsilon}\right) = s \lim_{t \rightarrow \infty} \frac{\varphi(t)}{t}, \quad s > 0.$$

4.2 Duality and Variational Formulas

We will show that the OCE emerges from a variational principle involving the φ -divergence. For that purpose, we need first to recall some results on conjugate functions, and convex integral functionals.

Let $\varphi \in \Phi$. The conjugate of φ , denoted by φ^* is defined by

$$\varphi^*(s) = \sup_{t \in \mathbb{R}} \{st - \varphi(t)\} = \sup_{t \in \text{dom } \varphi} \{st - \varphi(t)\} = \sup_{t \in \text{int dom } \varphi} \{st - \varphi(t)\}, \quad (4.33)$$

²The choice of $t^* = 1$ is for convenience. Any other point $t^* \in \text{int dom } \varphi$ can be picked.

where the last equality is from [26, Corollary 12.2.2]. The conjugate φ^* is a closed proper convex function, with $\text{int dom } \varphi^* = (a, b)$, where

$$a = \lim_{t \rightarrow -\infty} t^{-1} \varphi(t) \in [-\infty, +\infty); \quad b = \lim_{t \rightarrow +\infty} t^{-1} \varphi(t) \in (-\infty, +\infty].$$

Moreover, since φ is closed, its bi-conjugate $\varphi^{**} = \varphi$, (see e.g., [26]).

To establish the dual variational formula of the OCE we need to recall some results on normal convex integrands [29, Corollary 2E, p. 176]. Let $f : \mathbb{R} \times \Omega \rightarrow (-\infty, +\infty]$. If $f(\cdot, \omega)$ is (convex) and closed for almost all ω , and measurable in ω for each x such that $\text{dom } f(\cdot, \omega)$ has nonempty interior for every ω , then f is a normal (convex) integrand. The following result which allows to interchange integration and minimization is particularly useful for handling optimization problems with integral functionals. Here we state a special case of this result relevant to our applications, for a more general version see [29, Theorem 3A, p.185], [30, Theorem 14.60].

Theorem 4.1 *Let Ω be a σ -finite measure space, and let $\mathcal{X} := L^p(\Omega, \mathcal{F}, P)$, $p \in [1, +\infty]$. Let $g : \mathbb{R} \times \Omega \rightarrow (-\infty, +\infty]$ be a normal integrand, and define on $\mathcal{X}$ the integral functional $I_g(x) := \int_{\Omega} g(x(\omega), \omega) dP(\omega)$. Then,*

$$\inf_{x \in \mathcal{X}} \int_{\Omega} g(x(\omega), \omega) dP(\omega) = \int_{\Omega} \inf_{s \in \mathbb{R}} g(s, \omega) dP(\omega), \quad (4.34)$$

provided the lefthand side is finite. Moreover,

$$\bar{x} \in \underset{x \in \mathcal{X}}{\text{argmin}} I_g(x) \iff \bar{x}(\omega) \in \underset{s \in \mathbb{R}}{\text{argmin}} g(s, \omega), \text{ for a.e. } \omega \in \Omega. \quad (4.35)$$

We denote by $\mathcal{Q}$ the set of probability measures Q on $(\Omega, \mathcal{F})$ absolutely continuous with respect to P .

The following Theorem establishing the aforementioned dual representation of OCE was proved in Ben-Tal and Teboulle [7, Lemma 1], for the special case $\text{dom } \varphi = [0, +\infty]$. Essentially the same proof technique works for the more general case considered here.

Theorem 4.2 *Let $\varphi \in \Phi$, and let $X \in L^q$. Then,*

$$\inf_{Q \in \mathcal{Q}} \{I_{\varphi}(Q, P) + \mathbb{E}_Q(X)\} = \sup_{\eta \in \mathbb{R}} \{\eta - \mathbb{E}_P \varphi^*(\eta - X)\}. \quad (4.36)$$

Therefore, with $u(t) \equiv -\varphi^(-t)$, one has,*

$$S_u(X) = \inf_{Q \in \mathcal{Q}} \{I_{\varphi}(Q, P) + \mathbb{E}_Q(X)\}.$$

Proof. Let $\varphi \in \Phi$. We denote by v the optimal value of the left hand side minimization problem in (4.36). Fix $Q \ll P$, with density $z(\omega) := \frac{dQ(\omega)}{dP(\omega)}$. Using the definition of $I_{\varphi}(Q, P)$, (c.f. (4.31), (recalling that $\text{dom } \varphi = [\alpha, \beta]$), and (4.30), one thus have

$$v = \inf_{z \in L^p} \left\{ \int_{\Omega} \varphi(z(\omega)) dP(\omega) + \int_{\Omega} X(\omega) z(\omega) dP(\omega) \mid \int_{\Omega} z(\omega) dP(\omega) = 1, \alpha \leq z(\omega) \leq \beta, \text{ a.e.} \right\}.$$

The Lagrangian dual is given by,

$$\begin{aligned}
w : &= \sup_{\eta \in \mathbb{R}} \left\{ \eta + \inf_{\alpha \leq z(\cdot) \leq \beta} \left\{ \int_{\Omega} \varphi(z(\omega)) dP(\omega) - \int_{\Omega} (\eta - X(\omega)) z(\omega) dP(\omega) \right\} \right\} \\
&= \sup_{\eta \in \mathbb{R}} \left\{ \eta - \sup_{\alpha \leq z(\cdot) \leq \beta} \left\{ \int_{\Omega} (\eta - X(\omega)) dP(\omega) - \int_{\Omega} \varphi(z(\omega)) dP(\omega) \right\} \right\} \\
&= \sup_{\eta \in \mathbb{R}} \left\{ \eta - \int_{\Omega} \left[\sup_{\alpha \leq s \leq \beta} \{ (\eta - X(\omega)) s - \varphi(s) \} \right] dP(\omega) \right\} \\
&= \sup_{\eta \in \mathbb{R}} \left\{ \eta - \int_{\Omega} (\varphi)^*(\eta - X(\omega)) dP(\omega) \right\} = \sup_{\eta \in \mathbb{R}} \{ \eta - \mathbb{E}_P \varphi^*(\eta - X) \},
\end{aligned}$$

where the third equality follows by invoking Theorem 4.1, (note that since φ is convex with $\text{int dom } \varphi = (\alpha, \beta)$, it follows that the integrand in normal convex) and the last one from (4.33). Thus, w reduces to the desired expression in the right hand side of (4.36). It remains to show that $v = w$, i.e., there is no duality gap. For that, by invoking general convex duality results (see e.g., [11], [17]), one needs to verify that the following constraint qualification holds:

$$(CQ) \quad \exists \hat{z} \in L^p \text{ such that } \int_{\Omega} \hat{z}(\omega) dP(\omega) = 1, \alpha < \hat{z}(\omega) < \beta \text{ a.e.}$$

Since here we assumed that $\alpha < 1 < \beta$, then with $\hat{z}(\omega) = 1, a.e.$, (CQ) trivially holds and the proof is completed. $\square$

Theorem 4.2 provides a concrete realization of the representation theorem established by Föllmer and Shied [15, Theorem 6, p. 435], which we now recall.

Theorem 4.3 (*Representation Theorem of Föllmer-Shied [15]*).

Let $\mathcal{X} = L^\infty(\Omega, \mathcal{F}, P)$, and $\mathcal{Q}$ be the set of probability measures $Q \ll P$. Then any convex risk measure $\rho : \mathcal{X} \rightarrow \mathbb{R}$ can be represented via:

$$\rho(X) = \sup_{Q \ll P} \{ \mathbb{E}_Q(-X) - \alpha(Q) \}$$

for some closed convex "penalty function" $\alpha : P \rightarrow (-\infty, +\infty]$.

Comparing Theorem 4.2 and Theorem 4.3, we see that for the convex risk measure $\rho(X) = -S_u(X)$, the "penalty" function $\alpha(\cdot)$ is nothing else but the φ -divergence between the probability measures P and Q .

The next two results extend [9, Theorems 4.3, 4.2] to arbitrary distributions and gives an "inverse" dual representation of the φ -divergence in terms of the OCE; compare with Föllmer and Shied [15, Proposition 7, p. 436].

Theorem 4.4 Let $\varphi \in \Phi$ and set $u(t) = -\varphi^*(-t)$. Then, for any probability measure $Q \ll P$ one has,

$$I_\varphi(Q, P) = \sup \{ S_u(X) - \mathbb{E}_Q(X) \mid X \in L^q(\Omega, \mathcal{F}, P) \}.$$

Proof. Fix $Q \ll P$. Denote the value of the righthand side supremum by $R(Q, P)$. Using the definition of the OCE, with $u(t) = -\varphi^*(-t)$ one has:

$$\begin{aligned}
R(Q, P) &= \sup\{S_u(X) - \mathbb{E}_Q(X) \mid X \in L^q(\Omega, \mathcal{F}, P)\} \\
&= \sup_{X \in L^q} \left\{ \sup_{\eta \in \mathbb{R}} \{\eta - \mathbb{E}_P \varphi^*(\eta - X)\} - \mathbb{E}_Q(X) \right\} \\
&= \sup_{\eta \in \mathbb{R}} \left\{ \eta + \sup_{X \in L^q} \mathbb{E}_P \left\{ X \frac{dQ}{dP} - \varphi^*(\eta - X) \right\} \right\} \\
&= \sup_{\eta \in \mathbb{R}} \eta(1 - \mathbb{E}_P(\frac{dQ}{dP})) + \sup_{Y \in L^q} \mathbb{E}_P(Y \frac{dQ}{dP} - \varphi^*(Y)) \\
&= 0 + \sup_{Y \in L^q} \mathbb{E}_P(Y \frac{dQ}{dP} - \varphi^*(Y)).
\end{aligned}$$

Applying Theorem 4.1, the last supremum reduces to

$$\sup_{Y \in L^q} \mathbb{E}_P(Y \frac{dQ}{dP} - \varphi^*(Y)) = \mathbb{E}_P \left[\sup_{s \in \mathbb{R}} \{s \frac{dQ}{dP} - \varphi^*(s)\} \right] = \mathbb{E}_P \varphi^{**}(\frac{dQ}{dP}) = \mathbb{E}_P \varphi(\frac{dQ}{dP}),$$

where the last equality follows from the fact that φ being closed convex, one has $\varphi^{**} = \varphi$. Summarizing, we thus conclude that $R(Q, P) = \mathbb{E}_P \varphi(\frac{dQ}{dP})$, which is exactly (4.32). $\square$

The next result gives another formulation for the φ -divergence in terms of $S_u(\cdot)$.

Theorem 4.5 *Let $\varphi \in \Phi$ and $u(t) = -\varphi^*(-t)$. Then, for any probability measure $Q \ll P$ one has,*

$$I_\varphi(Q, P) = \sup\{S_u(X) \mid \mathbb{E}_Q(X) = 0 \mid X \in L^q(\Omega, \mathcal{F}, P)\}.$$

Proof. Fix $Q \ll P$. Using the definition of the OCE with $\eta - X := Y$, the supremum in the above right hand side can be written as

$$\sup_{\eta \in \mathbb{R}, Y \in L^q} \{\eta - \mathbb{E}_P \varphi^*(Y) \mid \mathbb{E}_Q(Y) = \eta\} = \sup_{Y \in L^q} \mathbb{E}_P(Y \frac{dQ}{dP} - \varphi^*(Y)).$$

Invoking Theorem 4.1, the desired result follows like at the end of the proof of Theorem 4.4. $\square$

The duality between the φ -divergence and the OCE is now used to complete the proof of Theorem 2.2(a), announced in Section 2.

Lemma 4.1 *Let u be an arbitrary but fixed utility function in $\mathbb{U}_0$, and let $\delta > 0$. For any random variable X with lower support $x_{\min} > -\infty$:*

$$\lim_{\delta \rightarrow \infty} S_\delta(X) = x_{\min},$$

where $S_\delta(X) = \sup_{\eta \in \mathbb{R}} \{\eta + \delta^{-1} \mathbb{E}_P u(\delta X - \delta \eta)\} = \delta^{-1} S_u(\delta X)$, (c.f. (2.14)).

Proof. Let $\varepsilon > 0$ be fixed, and let $\delta > 0$. Consider the convex optimization problem :

$$(P_\varepsilon) \quad v := \inf_{Q \in \mathcal{Q}} \{\delta^{-1} I_\varphi(Q, P) : \mathbb{E}_Q(X) \leq x_{\min} + \varepsilon\}.$$

Let $\lambda \in \mathbb{R}_+$ be the lagrange multiplier attached to the inequality constraint $\mathbb{E}_Q(X) \leq x_{\min} + \varepsilon$. Following the same arguments as in Theorem 4.2, one obtains that the Lagrangian dual of the above problem is:

$$\begin{aligned} v^* &= \sup_{\lambda \in \mathbb{R}_+} \left\{ -\lambda(x_{\min} + \varepsilon) + \sup_{\eta \in \mathbb{R}} \left\{ \eta + \frac{1}{\delta} \mathbb{E}_P(u(\delta(\lambda X - \eta))) \right\} \right\} \\ &= \sup_{\lambda \in \mathbb{R}_+} \left\{ -\lambda(x_{\min} + \varepsilon) + S_\delta(\lambda X) \right\}, \end{aligned}$$

where in the second equality we use the relation (2.14). Since (P_ε) is clearly feasible for any $\varepsilon > 0$, then from weak duality one has $v^* < +\infty$, and hence, (using again (2.14)):

$$\lim_{\lambda \rightarrow \infty} \lambda \left(\frac{S_\delta(\lambda X)}{\lambda} - (x_{\min} + \varepsilon) \right) \leq v^* < +\infty,$$

and hence $\lim_{\lambda \rightarrow \infty} \frac{S_\delta(\lambda X)}{\lambda} \leq (x_{\min} + \varepsilon)$. Therefore, with $\tau := \delta\lambda$, $S_\tau(X) = \frac{S_\delta(\lambda X)}{\lambda}$ and hence $\lim_{\tau \rightarrow \infty} S_\tau(X) \leq x_{\min} + \varepsilon$. Since $\varepsilon > 0$ was fixed but arbitrary, and in Theorem (2.2) we proved that $S_\tau(X) \geq x_{\min}$, the desired result follows. $\square$

4.3 Computing CVAR via The Dual Representation of the OCE

A popular coherent risk measure is the conditional value at risk we have already encountered in Section 2. The purpose of this section is to illustrate how some other well known representations of CVAR are also special cases of our dual formula given in Theorem 4.2.

Recall that CVAR is the negative of the OCE corresponding to the utility function (cf. Example 2.3),

$$u(t) = -\gamma \max\{0, -t\}, \quad (\gamma \equiv \gamma_2 > 1). \quad (4.37)$$

Its (concave) conjugate function is

$$u^*(s) = \begin{cases} 0 & \text{if } 0 \leq s \leq \gamma \\ -\infty & \text{otherwise,} \end{cases}$$

and so

$$\varphi(s) = -u^*(s) = \begin{cases} 0 & 0 \leq s \leq \gamma \\ \infty & \text{otherwise,} \end{cases} \quad (4.38)$$

namely the φ -divergence kernel is the indicator function of the closed interval $[0, \gamma]$. The next two examples illustrate two important special cases showing how CVAR can be computed from its dual representation given in terms of the φ -divergence.

Example 4.1 (Computing CVAR for discrete random variables)

In [28, Proposition 8], a formula for CVAR in the case of a discrete random variables was obtained. We show that this formula is a special case of the dual representation of $S_u(X)$. Let X be the random variable such that,

$$\Pr(X = x_i) = p_i \quad i = 1, \dots, n, \text{ where } x_1 \leq x_2 \leq \dots \leq x_n, \quad \sum_{i=1}^n p_i = 1, \quad p_i > 0.$$

By Theorem 4.2, the dual representation of $S_u(X)$ is

$$S_u(X) = \inf \left\{ I_\varphi(q, p) + \sum_{i=1}^n q_i x_i \mid \sum_{i=1}^n q_i = 1, \quad q \geq 0 \right\} \text{ with } I_\varphi(q, p) := \sum_{i=1}^n p_i \varphi(p_i^{-1} q_i).$$

For φ given by (4.38), the latter reduces to

$$S_u(X) = \inf \left\{ \sum_{i=1}^n q_i x_i \mid 0 \leq p_i/q_i \leq \gamma, \quad \sum_{i=1}^n q_i = 1, \quad q \geq 0 \right\}. \quad (4.39)$$

The solution of this “continuous knapsack” problem is well known: Let i^* be an index such that

$$\gamma \sum_{i=1}^{i^*} p_i \leq 1 < \gamma \sum_{i=1}^{i^*+1} p_i \quad (1 \leq i^* \leq n)$$

then the optimal solution of problem (4.39) is

$$\begin{cases} q_0^* = \gamma p_i & i = 1, 2, \dots, i^* \\ q_{i^*+1}^* = 1 - \gamma \sum_{i=1}^{i^*} p_i \\ q_j^* = 0, & j^* = i^* + 2, \dots, n. \end{cases}$$

Therefore, $S_u(X) = -\text{CVAR}_{1/\gamma}(X) = \gamma \sum_{i=1}^{i^*} (x_i - x_{i^*+1}) p_i + x_{i^*+1}$.

Example 4.2 (Computing CVAR for continuous random variables)

Let X be a continuous random variable with density $p(t)$ and support T . For u , and φ as in (4.37), (4.38), the dual representation of $S_u(X)$ is here³

$$S_u(X) = \inf_{q(\cdot)} \left\{ \int_T x(t) q(t) dt \mid 0 \leq q(t)/p(t) \leq \gamma, \quad q(t) \geq 0 \quad a.e., \quad \int_T q(t) dt = 1 \right\}. \quad (4.40)$$

Define $Q_\gamma = \{q(\cdot) \mid 0 \leq q(t)/p(t) \leq \gamma, \quad a.e\}$ and let λ^* be the Lagrange multiplier of the constraint

$$1 - \int_T q(t) dt = 0.$$

The Lagrangian of problem (4.40) is then

$$L(q(\cdot), \lambda) = \int_T x(t) q(t) dt - \lambda^* \int_T q(t) dt + \lambda^* = \int_T (x(t) - \lambda^*) q(t) dt + \lambda^*.$$

³This formula for CVAR is derived also in ([16, Theorem 4.39]).

The optimal density $q^*(\cdot)$ necessarily satisfies

$$\begin{aligned} q^*(t) &= \arg \min_{q \in Q_\gamma} L(q(\cdot), \lambda^*) \\ &= \arg \min_{q \in Q_\gamma} \int_T (x(t) - \lambda^*) q(t) dt = \begin{cases} 0, & \forall t \in T : x(t) > \lambda^* \\ \gamma p(t), & \forall t \in T : x(t) \leq \lambda^*. \end{cases} \end{aligned}$$

Now λ^* is determined from $\int_T q^*(t) dt = 1$, which is equivalent to

$$\gamma \int_{x(t) \leq \lambda^*} p(t) dt = 1 \iff \lambda^* = F^{-1}(1/\gamma), \quad (4.41)$$

where $F(x)$ is the distribution function of X . Substituting the optimal q^* in the objective function of (4.40), and using the relation (4.41), we get

$$\begin{aligned} S_u(X) &= \gamma \int_{x(t) \leq \lambda^*} x(t) p(t) dt = \\ &= \int_{x(t) \leq F^{-1}(1/\gamma)} x(t) p(t) dt \bigg/ \int_{x(t) \leq F^{-1}(1/\gamma)} p(t) dt \\ &= E(X \mid X \leq -\text{VAR}_{1/\gamma}(X)) = -\text{CVAR}_{1/\gamma}(X). \end{aligned}$$

5 Coherent/Convex Risk Measures and Other Related Certainty Equivalents

In this section we further study the risk measures associated with two certainty equivalents mentioned in the introduction, namely the classical certainty equivalent $C_u(\cdot)$ and the u-mean $M_u(\cdot)$, and we establish conditions under which these can be considered as coherent or/and convex risk measures. In particular, using our dual framework we show that the notion of bounded shortfall risk of Föllmer-Siied is in fact a special of the (negative) of the u-mean certainty equivalent, and we derive a dual representation given in terms of the φ -divergence, which recaptures the results developed in [15].

5.1 When is $C_u(X) = u^{-1}(Eu(X))$ a concave/coherent certainty equivalent

For a concave utility $u \in \mathbb{U}_0$, the classical certainty equivalent $C_u(X)$ is *not* necessarily concave, since u^{-1} is a *convex* function. The following theorem of Ben-Tal and Teboulle [6, Theorem 1, p. 1450] is the basis for characterizing those utilities for which $C_u(X)$ is concave.

Theorem 5.1 *Let $u \in C^3$, with $u' > 0$, $u'' < 0$, and let Z a random vector in $\mathbb{R}^n$. Consider the function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ given by*

$$f(y) = C_u(y^T Z) = u^{-1}(Eu(y^T Z)).$$

Then f is concave on $\mathbb{R}^n$ for any random variable Z , if and only if $1/r(t)$ is a concave function, where $r : \mathbb{R} \rightarrow \mathbb{R}$, defined by

$$r(t) = -\frac{u''(t)}{u'(t)}$$

is the Arrow-Pratt risk aversion measure (see [3], [25]). $\square$

A direct implication of Theorem 5.1 gives a characterization of the utilities for which $C_u(\cdot)$ is a concave certainty equivalent.

Corollary 5.1 *Let u be as in Theorem 5.1. Then $C_u(X)$ is a concave certainty equivalent, i.e., for all $0 \leq \lambda \leq 1$ and any random variables X_1, X_2 ,*

$$C_u(\lambda X_1 + (1 - \lambda)X_2) \geq \lambda C_u(X_1) + (1 - \lambda)C_u(X_2), \quad (5.42)$$

if and only if $1/r(\cdot)$ is concave.

Proof. By Theorem 5.1, the function $\theta(x, y) := C_u(xX_1 + yX_2)$ is concave. Thus, for all $0 \leq \lambda \leq 1$ and any vectors $(x_1, y_1) \in \mathbb{R}^2$, and $(x_2, y_2) \in \mathbb{R}^2$ one has:

$$C_u(\lambda[x_1X_1 + y_1X_2] + (1 - \lambda)[x_2X_1 + y_2X_2]) \geq \lambda C_u(x_1X_1 + y_1X_2) + (1 - \lambda)C_u(x_2X_1 + y_2X_2). \quad (5.43)$$

Thus, in particular, for the vectors $(x_1, y_1) = (1, 0)$, $(x_2, y_2) = (0, 1)$ inequality (5.43) reduces precisely to (5.42). $\square$

A wide class of utilities satisfying the requirement that $1/r(t)$ is concave is the class where $1/r(t)$ is *linear*, i.e.,

$$r(t) = \frac{1}{at + b} \quad (t > -b/a).$$

These utilities are in one of the following three forms:

$$u(t) = \begin{cases} b(1 - e^{-t/b}) & \text{if } a = 0, b \neq 0 \\ \log(b + t) & \text{if } a = 1 \\ (at + b)^{\frac{a-1}{a}} & \text{if } a \neq 0, a \neq 1. \end{cases}$$

Among all concave increasing utilities, the only one for which $C_u(\cdot)$ is shift-additive, is the first one, see [4]. Hence, we conclude that

$$\rho(X) = -C_u(X)$$

is a convex risk measure only for the class of exponential utilities

$$u_b(t) := b(1 - e^{-t/b}), \quad 0 < b \leq \infty. \quad (5.44)$$

Consequently, the only utility for which $-C_u(X)$ is a *coherent risk measure* (i.e., u is positively homogeneous and shift additive) is the *linear utility* $u(t) = t$ ($b \rightarrow \infty$ in (5.44)), in which case

$$\rho(X) = -C_u(X) = -\mathbb{E}(X).$$

5.2 The u-Mean Certainty Equivalent and Bounded Shortfall Risk

Based on the so-called “zero-utility” principle [12], the *u-mean certainty equivalent* $M_u(X)$ of a random variable X is defined by

$$\mathbb{E}_P u(X - M_u(X)) = 0. \quad (5.45)$$

If u is *strictly increasing*, then $M_u(X)$ is indeed uniquely defined by (5.45). A more general definition which allows for just *nondecreasing* utilities is

$$M_u(X) = \sup\{\eta \mid \mathbb{E}_P u(X - \eta) \geq 0\}. \quad (5.46)$$

The corresponding risk measure $-M_u(X)$ is the so-called *bounded shortfall risk*, see Föllmer and Schied [15]:

$$\rho_\ell(X) = \inf\{\eta \mid \mathbb{E}_P \ell(-X - \eta) \leq x_0\}. \quad (5.47)$$

The relation between M_u and ρ_ℓ is simply

$$\rho_\ell(X) = -M_u(X), \text{ where } \ell(t) = x_0 - u(-t). \quad (5.48)$$

Note that when u is concave and nondecreasing, then ℓ is convex and nondecreasing (a “penalty function”). Also $u(0) = 0$ if and only if $\ell(0) = x_0$ (“initial wealth”).

Now, the dual objective function in the problem of the right hand side of (5.46) is

$$h(\lambda) = \sup_{\eta \in \mathbb{R}} \{\eta + \lambda \mathbb{E}_P u(X - \eta)\} = S_{\lambda u}(X).$$

Clearly, $0 \notin \text{dom } h$, and hence the dual problem is

$$\inf_{\lambda > 0} \sup_{\eta \in \mathbb{R}} \{\eta + \lambda \mathbb{E}_P u(X - \eta)\},$$

which by standard duality for the concave problem (5.46) coincides with its primal right hand side. Therefore, we have

$$M_u(X) = \inf_{\lambda > 0} S_{\lambda u}(X). \quad (5.49)$$

From this relation, it immediately follows that

$$M_u(X) \leq S_u(X), \quad (5.50)$$

and that $M_u(\cdot)$ is *concave*. Note that the inequality (5.50) implies that a decision-maker using the u -mean CE is more risk averse than one using the OCE.

Example 5.1 Consider the utility function $u(t) = -\gamma \max\{0, -t\}$ with $\gamma > 1$, for which $-S_u(X) = \text{CVAR}_{1/\gamma}(X)$. For this utility function, it follows easily from (5.46) that $M_u(X) = x_{\min}$, (the left support of X), which for certain random variables can be much smaller than CVAR.

A further interesting implication of the relation (5.49) is obtained by invoking the dual representation of the OCE given in Theorem 4.2. Indeed, since $(\lambda u)^*(t) = \lambda u^*(\lambda^{-1}t)$, then with $\hat{\varphi}(t) := \lambda \varphi(\lambda^{-1}t)$, it follows that,

$$\begin{aligned} S_{\lambda u}(X) &= \inf_{Q \in \mathcal{Q}} \{\mathbb{E}_Q(X) + I_{\hat{\varphi}}(Q, P)\} \\ &= \inf_{Q \in \mathcal{Q}} \{\mathbb{E}_Q(X) + \mathbb{E}_P(\lambda \varphi(\frac{dQ}{\lambda dP}))\}, \end{aligned}$$

and thus by (5.49),

$$M_u(X) = \inf_{\lambda > 0} \inf_{Q \in \mathcal{Q}} \{ \mathbb{E}_Q(X) + \mathbb{E}_P(\lambda \varphi(\frac{dQ}{\lambda dP})) \}.$$

Now, since $\ell(t) = x_0 - u(-t)$, and $\varphi(t) = -u^*(t)$, one easily verifies that $\varphi(t) = x_0 + \ell^*(t)$, and hence the last formula for M_u reads,

$$M_u(X) = \inf_{\lambda > 0} \inf_{Q \in \mathcal{Q}} \{ \lambda x_0 + \mathbb{E}_Q(X) + \mathbb{E}_P(\lambda \ell^*(\frac{dQ}{\lambda dP})) \},$$

which recovers a result of Föllmer and Schied [15, Theorem 10].

We have already seen that $\rho(X) = -M_u(X)$ is a convex risk measure.

We end this section by characterizing the class of utilities for which $-M_u(\cdot)$ is a coherent risk measure, which happens to coincide with the same class of piecewise linear utilities (c.f. Example 2.3), under which $-S_u(\cdot)$ is coherent.

Proposition 5.1 *In the class $\mathbb{U}_0^<$ of strongly risk averse utilities, $\rho(X) = -M_u(X)$ is coherent if and only if u is a piecewise linear function of the form*

$$u(t) = \begin{cases} \gamma_1 t & t \geq 0 \\ \gamma_2 t & t < 0 \end{cases} \quad \text{where } 0 \leq \gamma_2 < 1 < \gamma_1.$$

Proof. We first show that if $M_u(X)$ is homogeneous, then u is homogeneous, (the reverse implication is obvious from (5.46)). Consider the random variable $\bar{X}_0$:

$$\Pr(\bar{X}_0 = \alpha) = p_0, \quad \Pr(\bar{X}_0 = \beta) = 1 - p_0$$

where

$$\alpha < 0 < \beta \quad \text{and} \quad p_0 = \frac{u(\beta)}{u(\beta) - u(\alpha)}.$$

Note that since $u \in \mathbb{U}_0^<$, then $0 < p_0 < 1$ and $M_u(\bar{X}_0) = 0$.

Consider the case when $\lambda \in (0, 1)$. Then, $M_u(\lambda \bar{X}_0)$ is determined by the equation:

$$\begin{aligned} 0 &= p_0 u(\lambda \alpha - M_u(\lambda \bar{X}_0)) + (1 - p_0) u(\lambda \beta - M_u(\lambda \bar{X}_0)) \\ &= p_0 u(\lambda \alpha) + (1 - p_0) u(\lambda \beta), \quad (\text{since } M_u(\lambda \bar{X}_0) = \lambda M_u(\bar{X}_0) = 0) \\ &\geq p_0 \lambda u(\alpha) + (1 - p_0) \lambda u(\beta), \quad (\text{by concavity of } u, \text{ and } u(0) = 0) \\ &= \lambda [p_0 u(\alpha) + (1 - p_0) u(\beta)] = 0 \quad (\text{by definition of } p_0). \end{aligned}$$

Hence, equalities hold in the above relations, which implies in particular

$$p_0 [u(\lambda \alpha) - \lambda u(\alpha)] + (1 - p_0) [u(\lambda \beta) - \lambda u(\beta)] = 0.$$

But, since $u \in \mathbb{U}_0^<$ is concave, each term in the squared brackets is nonnegative, and so

$$\begin{aligned} u(\lambda \alpha) &= \lambda u(\alpha) & (\alpha < 0) \\ u(\lambda \beta) &= \lambda u(\beta) & (\beta > 0), \end{aligned}$$

so that for any $\lambda \in (0, 1)$, one has proved that $u(\lambda t) = \lambda u(t) \quad \forall t$. Likewise, the case $\lambda > 1$ can be handled similarly as in the proof of Proposition 3.1, and this complete the proof that u is homogeneous. Finally, by Lemma 3.1, u must have the claimed piecewise linear form. $\square$

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